1000000 raw units can mean 1 token

Frontend dApp Engineer · 6-decimal example · Five answers

FE-F-02 | Foundation | Practical

How do you format a number from a `uint256` value, like an ERC-20 token balance, for display in a UI?

Answer

For a token with 6 decimals, viem formatUnits(1000000n, 6) returns '1'.

Reading chain state and authorizing actions

FE-F-01 Foundation Knowledge

What is a"provider" in Ethers.js or Viem and what is its role?

Answer

In ethers v6, a Provider reads chain data; a Signer authorizes transactions. In viem, Public Client exposes reads and Wallet Client handles signing via a wallet or local account.

Format units without losing integer precision

FE-F-02 Foundation Practical

How do you format a number from a `uint256` value, like an ERC-20 token balance, for display in a UI?

Answer

Use formatUnits(balance, decimals), keeping balance as bigint. In viem, formatUnits(1000000n, 6) returns '1'. Get decimals for that token and chain; never assume 18.

ERC-20 decimals() is optional. If absent, use verified token metadata or label raw units instead of silently assuming 18.

A cancellation races the pending transaction

FE-I-01 | Intermediate | Practical

A user reports their transaction is"stuck" as pending. What are the possible causes and how could you handle this in the UI?

Answer

Low fees or an earlier pending nonce can block it. Show pending status. Speed-up uses the same nonce with higher fees; cancellation competes with the original and can fail.

NFT ownership needs an index

FE-A-01 | Advanced | Architecture

How would you design a frontend to display a user's entire NFT collection, including images and metadata, across multiple contracts?

Answer

Query an ownership indexer by wallet and chain, paginate results, then render token IDs, metadata and images. Handle missing media. balanceOf cannot list every collection.

Typed signatures still need replay controls

FE-E-01 Expert Design

You are building a dApp that requires users to sign messages for off-chain actions (e.g., voting, placing orders on an off-chain order book). Describe the EIP-712 standard and explain why it is superior to signing a simple string with `personal_sign`.

Answer

It signs typed fields with a domain, so wallets can show structured terms. Include chainId and verifyingContract. The app must still enforce nonces and expiry.